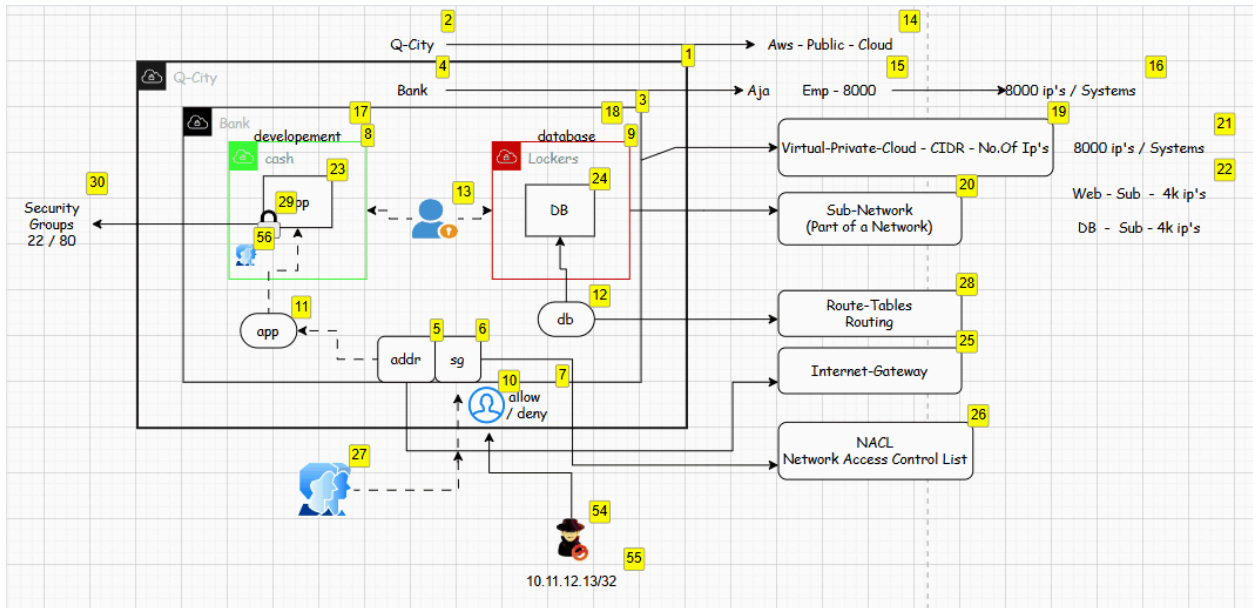




Set up a custom VPC with:

- 1 Public Subnet
- 1 Private Subnet
- Internet Gateway (IGW)
- Route Tables (RTs)
- Attach IGW to the VPC
- Route internet traffic from the public subnet

Lab Requirements

- AWS Account
- IAM User with VPC full access
- Region: `us-east-1` (or your preferred region)

Step 1: Create a Custom VPC

1. Go to the **VPC Dashboard**:
Console > VPC > Your VPCs > Create VPC
2. Select **VPC only**.
3. Fill in the details:
 - **Name tag:** `MyLabVPC`
 - **IPv4 CIDR block:** `10.0.0.0/16`
 - Leave the rest as default
 - Click **Create VPC**

Step 2: Create Internet Gateway (IGW)

1. In the **VPC Dashboard**, go to **Internet Gateways > Create Internet Gateway**
2. Name tag: `MyLabIGW`
3. Click **Create Internet Gateway**

4. Select the IGW and click **Attach to VPC**

- Choose **MyLabVPC**
-

Step 3: Create Subnets

3.1 Public Subnet

1. Go to **Subnets > Create Subnet**

2. Choose:

- **VPC:** **MyLabVPC**
- **Subnet name:** **MyLabPublicSubnet**
- **Availability Zone:** **us-east-1a** (or your choice)
- **CIDR block:** **10.0.1.0/24**

3. Click **Create Subnet**

3.2 Private Subnet

1. Again, click **Create Subnet**

2. Choose:

- **VPC:** **MyLabVPC**
- **Subnet name:** **MyLabPrivateSubnet**
- **Availability Zone:** **us-east-1a** (same AZ for simplicity)
- **CIDR block:** **10.0.2.0/24**

3. Click **Create Subnet**
-

Step 4: Create Route Tables

4.1 Public Route Table

1. Go to **Route Tables > Create Route Table**
2. Name: **MyLabPublicRT**
3. VPC: **MyLabVPC**
4. Click **Create**

Associate Public Subnet:

1. Select **MyLabPublicRT > Subnet Associations > Edit subnet associations**
2. Select **MyLabPublicSubnet** and click **Save**

Add Internet Route:

1. Select **MyLabPublicRT > Routes > Edit routes**
2. Click **Add route**
 - **Destination:** **0.0.0.0/0**
 - **Target:** Select your **Internet Gateway** (MyLabIGW)
3. Click **Save routes**

4.2 Private Route Table (Optional for now)

You can keep the default route table for private subnet with **no internet access**, or:

- Create a new route table named **MyLabPrivateRT**
- Associate it with **MyLabPrivateSubnet** (no IGW route)

Validation Checklist

Resource	Name	CIDR / Notes
VPC	MyLabVPC	10.0.0.0/16
Internet Gateway	MyLabIGW	Attached to VPC
Public Subnet	MyLabPublicSubnet	10.0.1.0/24
Private Subnet	MyLabPrivateSubnet	10.0.2.0/24
Public Route Table	MyLabPublicRT	IGW route + association
Private Route Table	MyLabPrivateRT	Local route only
